

Clinical Presentation

- **Figure 178-5:** Late stage of **staphylococcal scalded-skin syndrome (SSSS)** showing generalized desquamation with shedding of large sheets of skin.
- Characteristic findings include: - Widespread erythema - Superficial blisters - Desquamation of large skin sheets - Superficial erosions, particularly around the eyes and face - Crusting and denuded skin in affected areas, as seen in a 10-month-old child

Staphylococcal-Mediated Skin Eruptions

- **Bullous impetigo:**
- Localized form of *Staphylococcus aureus* infection
- Presents with regional bullae and denuded skin
- May or may not culture positive for *S. aureus*
- **Scarlatiniform variant:**
- Possibly an abortive form of SSSS
- Resembles early erythrodermic and late desquamative stages
- Minimal or no blistering
- Also known as **staphylococcal scarlet fever**
- **Overlap syndromes:**
- *S. aureus* strains can produce both exfoliative toxins (ETs) and superantigenic toxins
- ETs may have superantigenic properties
- Rare cases may show features of both **toxic shock syndrome** and **SSSS**

Diagnosis and Differential Diagnosis

- **Histopathology:** - Intraepidermal cleavage beneath and within the **stratum granulosum** - Presence of acantholytic cells in the cleavage

space - Minimal inflammation in epidermis and dermis (except in bullous impetigo, which shows more neutrophils)

- **Key differential: Toxic Epidermal Necrolysis (TEN):** - SSSS typically affects **neonates and young children**, while TEN is more common in adults (though overlap exists) - **Mucosal involvement:** - Absent in SSSS - Present in TEN - **Diagnostic tools:** - **Tzanck smear:** Shows acantholytic cells in SSSS, not in TEN - **Frozen section biopsy:** - SSSS: Superficial subgranular acantholysis - TEN: Full-thickness epidermal necrosis and dermal-epidermal separation

Treatment and Prognosis

- **Bullous impetigo:**
 - Topical **mupirocin ointment** alone or with oral antibiotics
 - Excellent prognosis
- **SSSS:**
 - Requires hospitalization
 - **Intravenous anti-staphylococcal antibiotics** (e.g., nafcillin, oxacillin, or vancomycin if MRSA suspected)
 - Transition to oral antibiotics after clinical improvement
 - **Supportive care:**
 - Fluid and electrolyte management
 - Maintenance of skin barrier function
 - Use of **non-adherent dressings** (e.g., petrolatum-impregnated gauze)
 - **Incubators** for neonates to regulate temperature and humidity
- **Adjunctive therapy:**
 - Topical mupirocin to impetiginized areas, including primary infection sites
- **Complications:**
 - Fluid and electrolyte imbalances due to skin barrier disruption
 - Sepsis is uncommon in children

- **Mortality:**
- **Children:** Very low ($\sim 2\%$), rarely due to sepsis
- **Adults:** Higher ($\sim 10\%$), often due to underlying comorbidities and increased risk of sepsis